

High-burnup structure formation as a second-order phase transition

A Landau functional for SCIAANTIX

iHighBurnupStructureFormation = 4

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SCIANTIX status

The model

Results

Model applied to SCIANTIX

SCIANTIX status

SCIANTIX had three formation models for the restructured volume fraction

option	form	driven by
1	KJMA, Barani (2020)	effective burnup
2	KJMA + $bu_{\text{inc}} = 15$, Biswas & Aagesen (2025)	effective burnup
3	KJMA on ρ_d , Veshchunov & Shestak (2009)	local burnup

All three are **empirical transformation kinetics**: a fitted sigmoid in burnup. None of them produces the **mean misorientation** Θ of the subgrains, and the **subgrain radius** r_n .

The data this model is built on

EBSD — Zacharie-Aubrun *et al.* (2022), Onofri *et al.* (2025). 42 radial positions on 7 rods, with TRANSURANUS local conditions.

quantity	N	what it is
mean misorientation Θ	41	$[A_{\text{Mis}}(f_1 - f_{10}) + 10f_{10}]/100$
restructured fraction X	27	f_{10} , area fraction above 10°
subgrain radius r_n	14	ECD50 % /2

The model

HBS as a second-order phase transition

Order parameter

$$\eta = \frac{\theta}{\theta_{\max}}, \quad \theta_{\max} \equiv \theta_{\text{HAGB}} = 10^\circ$$

the mean misorientation normalised to the LAGB/HAGB boundary.

External condition: the local burnup, entering *only* through the dislocation density

$$\rho_{\text{tot}}(bu) = 10^{2.2 \times 10^{-2} bu + 13.8}$$

Nogita & Une (1994)

Free energy

$$F(bu, \eta) = C_0 + C_2 \eta^2 + C_4 \eta^4$$

Only even powers: the energy cannot depend on the *sign* of a misorientation.

Minimising F gives Θ ; the same wall geometry gives r_n ; the lever rule gives X .

Three adjustable parameters

β (wall geometry), k (sweeping), ρ_c (strain-field cut-off).

The geometric wall

A low-angle boundary *is* a wall of dislocations. Two classical relations fix the whole geometry.

1. Spacing sets the angle $\theta = b/d$

so a wall of n dislocation families carries a line length per unit *wall area*

$$L_A = n\theta/b$$

Close the two on each other:

$$\rho_{\text{LAGB}} = L_A \cdot S/V$$

$$\Rightarrow \sqrt{\rho_{\text{LAGB}}} = \frac{3n\theta}{\beta b}$$

2. Wall area per unit volume

$$S/V = 3\sqrt{\rho_{\text{LAGB}}} / \beta$$

β : the one geometric parameter (Rest & Hofman 2000).
 $n = 2$, Gourdet & Montheillet, not fitted (between 1-3).

Everything follows from θ alone

$$\rho_{\text{LAGB}} = \left(\frac{3n\theta}{\beta b} \right)^2, \quad r_n \simeq \frac{\beta^2 b}{6n\theta}$$

The dislocation balance

Every dislocation is in exactly one of three states, and they must add up to ρ_{tot} :

$$\underbrace{\rho_{\text{ord}} = \rho_{\text{LAGB}}^{\text{max}} \eta^2}_{\text{condensed into low-angle walls}} + \underbrace{\rho_{\text{swept}} = (\rho_{\text{tot}} - \rho_{\text{ord}}) \frac{\Delta R}{R}}_{\text{annihilated by moving boundaries}} + \underbrace{\rho_{\text{free}}}_{\text{still a random array}} = \rho_{\text{tot}}$$

Each population carries the line energy of the state it is in:

$$F = \rho_{\text{free}} A_1 G b^2 + \rho_{\text{ord}} A_2 G b^2, \quad A_i = \frac{f(\nu)}{4\pi} \ln \frac{R_i}{b}$$

free ones screened at ρ_c , wall ones by the wall itself at ρ_{tot} ; the swept ones carry nothing.

$$C_0 = \rho_{\text{tot}} A_1 G b^2, \quad C_4 = \rho_{\text{LAGB}}^{\text{max}} \left. \frac{\Delta R}{R} \right|_{\text{max}} A_1 G b^2$$
$$C_2 = \underbrace{\rho_{\text{LAGB}}^{\text{max}} (A_2 - A_1) G b^2}_{\text{walls: the screening gain}} - \underbrace{\rho_{\text{tot}} \left. \frac{\Delta R}{R} \right|_{\text{max}} A_1 G b^2}_{\text{sweep}}$$

The engine of the transition is $A_2 - A_1$. As burnup rises, ρ_{tot} rises, the screening length inside the wall shrinks, A_2 falls below A_1 — and it becomes cheaper to store dislocations in walls than to leave them random.

Equilibrium

$$\frac{\partial F}{\partial \eta} = 0 \implies \eta^2 = -\frac{C_2}{2C_4}, \quad \text{clipped at 0}$$

$$\eta = 0 \text{ wherever } C_2 \geq 0.$$

Condition on the minimisation problem: dislocation balance

The walls cannot hold more dislocations than exist, so $\rho_{\text{ord}} \leq \rho_{\text{tot}}$. The equilibrium is the minimum of F on the **admissible interval**, not the free stationary point:

Eq. (7b) — admissibility

$$\eta_{\text{bal}} = \sqrt{\min\left(\frac{\rho_{\text{tot}}}{\rho_{\text{LAGB}}^{\text{max}}}, 1\right)}, \quad \eta_{\text{eq}} = \min\left(\sqrt{-\frac{C_2}{2C_4}}, \eta_{\text{bal}}\right)$$

Results

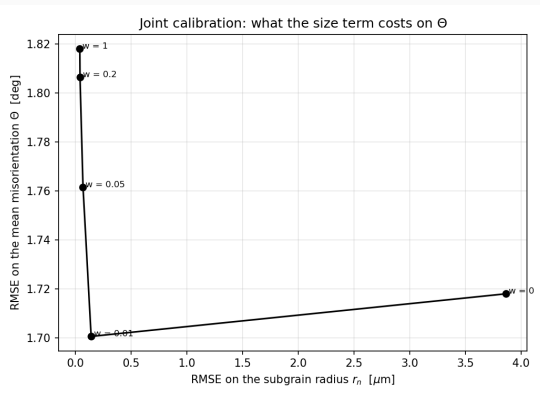
Calibration

Joint fit of β, k, ρ_c on Θ and the subgrain sizes, while the restructured fraction is *not* in the objective.

$$J = \frac{\langle \Delta \Theta^2 \rangle}{\text{var } \Theta} + w \frac{\langle \Delta r^2 \rangle}{\text{var } r}$$

dimensionless and symmetric;
differential_evolution, 6 seeds, all
converging to $J = 0.2398$.

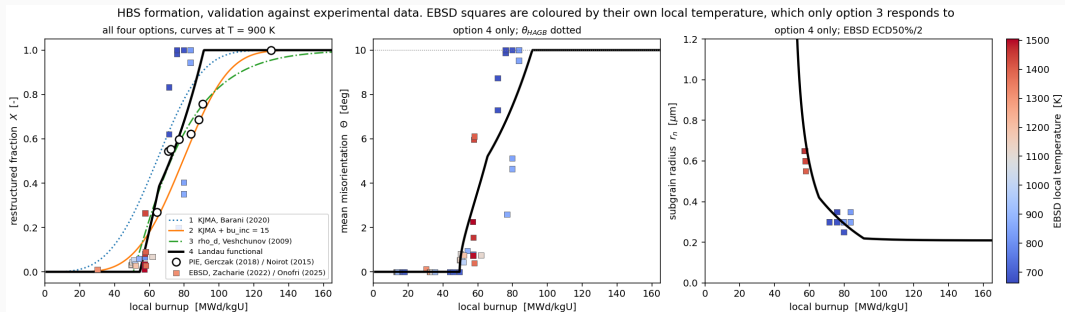
n (not fitted)	2
β	33.547
k	0.04697
ρ_c	$1.166\text{e}15 \text{ m}^{-2}$



What the balance bought

	RMSE / R^2
Θ ($N = 41$, calibrated)	1.7616° / 0.772
X ($N = 27$, <i>not</i> fitted)	0.2053 / 0.728
r_n ($N = 14$, calibrated)	$0.0678 \mu\text{m}$ / 0.774
PIE fraction ($N = 8$, <i>out-of-sample</i>)	0.1298 / +0.553

Validation against both datasets



Against the three models SCIANTIX already had

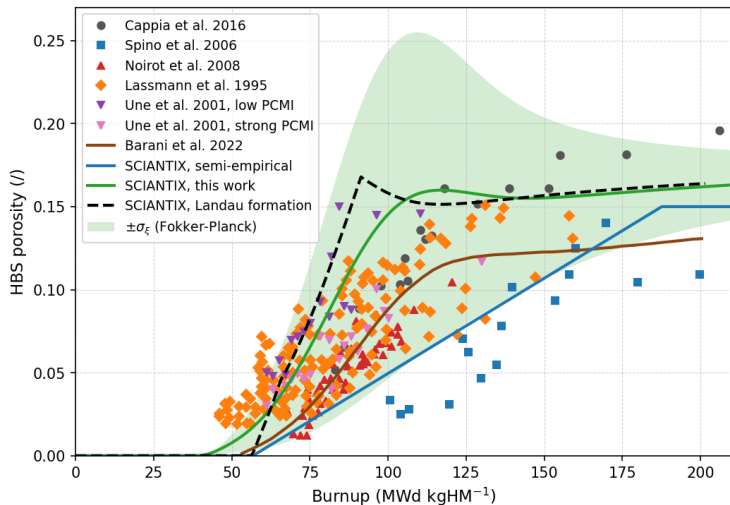
Same history, same initial conditions, same time stepping; only the formation model varies.
Burnup [MWd/kgU] at which α_r first reaches:

threshold	1 KJMA	2 KJMA+ bu_{inc}	3 ρ_d	4 Landau
1 %	19.5	34.5	51.2	54.9
10 %	37.8	52.8	54.8	57.5
50 %	64.2	79.3	72.3	71.8
90 %	90.1	105.2	112.3	88.2
99 %	109.7	124.6	161.7	91.1

- Through the onset, option 4 sits **within 0.5 MWd/kgU of option 3** at the 50 % crossing (an independently calibrated model, so a genuine cross-check).
- They part at the top: the lever rule **saturates** when Θ hits the 10° cap, whereas KJMA approaches 1 asymptotically.

Model applied to SCIANTIX

Does the new formation model disturb the porosity model?



Largest deviation $+0.045$
absolute at
91.3 MWd/kgHM
(+37 %), exactly the
saturation burnup.
Elsewhere: ≤ 0.017
below 75, ≤ 0.008 above
120, and the two end
states are **0.9 % apart**.

Summary

Backup — the equations

$$(1) \rho_{\text{tot}}(bu) = 10^{2.2 \times 10^{-2} bu + 13.8}$$

$$(2) G(T, P, x, q), \nu(T, P, x, q) \quad \text{NEA/NSC/R(2024)1 p. 124; } f(\nu) = \frac{1-\nu/2}{1-\nu}$$

$$(3) \rho_{\text{LAGB}}^{\text{max}} = \left(\frac{3n\theta_{\text{max}}}{\beta b} \right)^2, \quad (S/V)_{\text{max}} = \frac{9n\theta_{\text{max}}}{\beta^2 b}, \quad \left. \frac{\Delta R}{R} \right|_{\text{max}} = \frac{k \rho_{\text{LAGB}}^{\text{max}}}{\rho_{\text{tot}}}$$

$$(4) A_1 = \frac{f(\nu)}{4\pi} \ln \frac{\rho_{\text{c}}^{-1/2}}{b}, \quad A_2 = \frac{f(\nu)}{4\pi} \ln \frac{\rho_{\text{tot}}^{-1/2}}{b}$$

$$(5,6) F = \rho_{\text{free}} A_1 G b^2 + \rho_{\text{ord}} A_2 G b^2 = C_0 + C_2 \eta^2 + C_4 \eta^4$$

$$(7,7b) \eta_{\text{eq}} = \min(\sqrt{-C_2/2C_4}, \eta_{\text{bal}}), \quad \eta_{\text{bal}} = \sqrt{\min(\rho_{\text{tot}}/\rho_{\text{LAGB}}^{\text{max}}, 1)}$$

$$(8) \Theta = \min(\eta_{\text{eq}} \theta_{\text{max}} \cdot \frac{180}{\pi}, \theta_{\text{HAGB}})$$

$$(9) r_n = \min\left(\frac{1.5}{(S/V)}\left(1 + \frac{\Delta R}{R}\right), R_{\text{grain}}\right)$$

$$(10) X = \text{clip}\left(\frac{\Theta - \theta_{\text{u}}}{\theta_{\text{HAGB}} - \theta_{\text{u}}}, 0, 1 - 10^{-9}\right)$$

Backup — regimes and parameters

Regimes (T -, P -, x -independent)

condition	b_u [GWd/tU]
$C_2 = 0$, Θ leaves 0	49.56
$\Theta = \theta_u$, X leaves 0	54.56
$\Theta = \theta_{\text{HAGB}}$	91.32

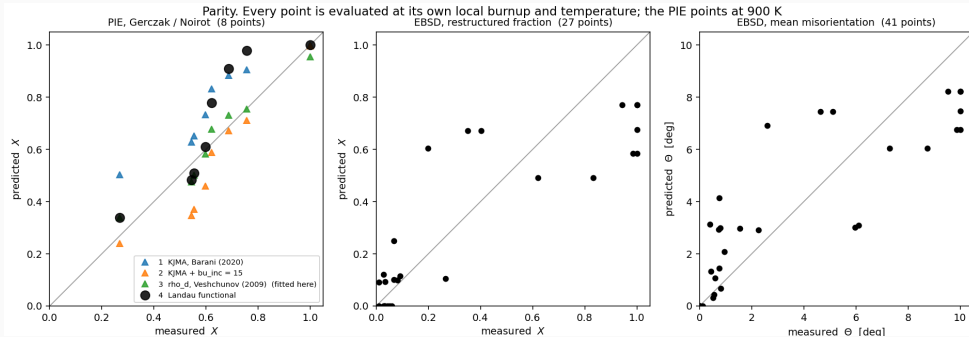
The balance bound sets η over roughly **65.8–91.2 GWd/tU**, and at 11 of the 41 EBSD points.

Fixed constants

b	3.889e-10 m
θ_{max}	$= \theta_{\text{HAGB}}$
θ_{HAGB}	10°
θ_u	2.20° (measured)
n	2

θ_u is the median A_{Mis2Mean} over the 27 points carrying a restructured fraction — a measured quantity, not a fitted one. An independent fit gives 1.55° , inside the measured range 0.9–6.1°.

Backup — parity plots



Backup — the four formation options in a full run

HBS formation options on the same irradiation, porosity model 3 throughout

